4.2 Stoichiometry

Problems Worksheet

1. Hydrogen sulfide reacts with zinc oxide to produce water and zinc sulfide.
   a. Write an equation for the reaction.

   In an experiment, 8.90 g of zinc oxide reacts.

   b. Determine the number of moles zinc sulfide produced.

   c. Determine the mass of hydrogen sulfide required.

   d. Determine the mass of water produced.

2. A precipitate of lead(II) iodide is produced when lead(II) nitrate reacts with sodium iodide.
    a. Write an equation for the reaction.

    b. Determine the mass of lead(II) iodide produced if 9.70 g of lead(II) nitrate reacts with excess sodium iodide.

3. Iron ore is comprised mostly of the mineral haematite, which is iron(III) oxide (Fe_2O_3). The iron is extracted from the ore by a reaction with carbon monoxide in a blast furnace. The overall equation for the process is shown:

    $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

    a. In WA, approximately 2.26 tonnes of haematite is mined each day. Calculate the mass of iron which could be produced from this ore.

    b. What mass of carbon monoxide would be required to extract the iron?

    c. What mass of carbon dioxide would be produced?

4. Sulfuric acid is made by a series of chemical reactions in the Contact Process. The equations are:

$S + O_2 \rightarrow SO_2$
$2SO_2 + O_2 \rightarrow 2SO_3$
$SO_3 + H_2O \rightarrow H_2SO_4$

a. The engineers at a sulfuric acid plant want to produce 1.00 tonnes of sulfuric acid per day. Calculate the mass of sulfur trioxide they will need to produce each day.

b. What mass of sulfur will they need?

c. Extension: combine the three equations into one overall equation for the process.

5. In a submarine, aircraft or spacecraft, oxygen can be generated in an emergency with an ‘oxygen candle’ containing sodium perchlorate:

$2NaClO_3(s) \rightarrow 2NaCl(s) + 3O_2(g)$

a. Determine the mass of oxygen gas produced by 1.000 kg of sodium perchlorate.

b. What mass of sodium chloride will be produced in this reaction?

An alternative oxygen candle uses the following reactin:

$LiClO_4(s) \rightarrow LiCl(s) + 2O_2(g)$

c. Determine whether this candle would be more or less effective at producing oxygen than the sodium perchlorate candle. Use a calculation to justify your answer.

6. Sodium chloride can be produced in an explosive reaction between sodium metal and chloride gas. In an experiment, 43.0 g of sodium is reacted with 75.1 g of chlorine gas.
    a. Determine the limiting reagent.

    b. Determine the mass of sodium chloride produced.

    c. What mass of the excess reagent will be left over after the reaction?

7. Sodium nitrite can react with potassium permanganate and hydrochloric acid according to the following equation:

$5NaNO_2(aq) + 2KMnO_4(aq) + 6HCl(aq) \rightarrow 5NaNO_3(aq) + 2KCl(aq) + 2MnCl_2(aq) + 3H_2O(l)$

Determine the mass of manganese chloride produced if 5.00 g of $NaNO_2$ reacts with 8.50 g of HCl.